

INTEGRATION AND INTEROPERABILITY OF NATIONAL IDENTIFICATION SYSTEMS:

A study of Ghana's National Data Governance situation and its impact on cross-border data flow.

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List Of Abbreviation

ACFTA	African Continental Free Trade Area
MNO	Mobile Network Operators
MMP	Mobile Money Providers
ENBIC	ECOWAS National Biometric Identity Card
AU	African Union
ECOWAS	Economic Community of West African States
e-GIF	Electronic Government Interoperability Framework
NIA	National Identification Authority
EC	Electoral Commission
DVLA	Driver and Vehicle Licensing Authority
NHIA	National Health Insurance Authority
DGI	Data Governance Institute
ICAO	International Civil Aviation Organisation
eIDAS	Electronic Identification, Authentication, and trust Services
ID	Identity

1.0 Introduction

Data plays a pivotal role in development planning by providing critical insights into population needs, guiding intervention strategies, and enabling the evaluation of outcomes. National identification systems serve as an essential data source for governments, facilitating access to fundamental rights, services, opportunities, and protections.

Over the past decade, African governments have made significant investments in developing national identification data systems. However, these efforts have often been fragmented, leading to challenges in cross-border data flow due to non-standardised, localised regulations.

This fragmentation hampers the movement of people and the seamless conduct of business across borders. This study delves into Ghana's data governance framework within the broader context of the African Union's policy framework, examining the integration and interoperability of the country's national identification systems and their implications for cross-border Data flow.

1.1 Background

After many years of using the national passport and voter's identity card to verify and validate citizens' legal identity, the Ghana government created the National Identification Authority to issue the "Ghana card" as the sole identification document for citizens and non-citizens. Act 707 established the National Identification Authority to create, maintain, provide, and promote the use of national identity cards to advance economic, political, and social activities in the country.¹ The National Identity Register, as amended by Act 950, governs the data collected and stored by the Authority in issuing the national identity card.² Aside from the National Identification Authority, four other government agencies also issue identification cards. They include the Passport Office, Electoral Commission, Driver and Vehicle License Authority, and Social Security and National Insurance Trust. It appears all of these identification issuing agencies operate in silos with no technical system integrations that enable interoperable data flow for identity verification and validation within or across borders. Interoperability is the ability of different functional units—e.g., systems, databases, devices, or applications—to communicate, execute programs, or transfer data in a manner that requires the user to have little or no knowledge of those functional units (ISO/IEC 2382).³ This can only be possible with a data governance framework that standardises data format, IT infrastructure, and codes of conduct that enable ongoing, collaborative, operational decision-making about data use.⁴ This appears to be absent in Ghana's case. The country, however, has a draft data-sharing policy and data protection in place to create an enabling environment for the sharing and use of government data for socioeconomic development and a framework that addresses in detail the processing and retention of personal data peculiar to a data system.

The absence of an effective data governance framework creates inconsistencies in the various data systems of Ghana's Personal Identification issuers. For example, a data subject's information captured by the Electoral Commission may be in different formats from what is captured on the same data subject by the National Identification Authority. Adopting data governance best practices will help Ghana make the most of its national identification systems and avoid operational or analytic issues that result from data inconsistencies.⁵ A fragmented identification data system may hinder regional data integration efforts such as the one by the African Union through the AU Data Policy Framework. This study contributes to knowledge and informs local policy action towards an integrated Pan-African National Identification system.

1. Laws of Ghana. (n.d.). National Communications Authority. Retrieved from

<https://nca.org.gh/wp-content/uploads/2020/09/National-Identification-Authority-Act-2006-Act-707.pdf>

2. National Identification Register (Amendment) Act, 2017 (n.d.). Ghana Business Regulation Reform. Retrieved from https://www.brr.gov.gh/reg_details.php?id=Njk3

3. International Standard ISO/IEC 2382. (n.d.) IEC Webstore. Retrieved from https://webstore.iec.ch/preview/info_isoiec2382%7Bed1.0%7Den.pdf

4. Lucas, S. (2022). From data revolutionary to data governance advocate and back again: Three lessons from the journey. Hewlett Foundation. Retrieved from <https://hewlett.org/from-data-revolutionary-to-data-governance-advocate-and-back-again-three-lessons-from-the-journey>

5. Harrington, D. (n.d.). What Is Data Governance? Framework and Best Practices. Varonis. Retrieved from <https://www.varonis.com/blog/data-governance>

1.2 Objective of research

The research objective is to assess Ghana's data governance framework within the conceptual framework of the AU Data Policy Framework, focusing mainly on integration and interoperability (refer to 5.4.3; page 51), particularly of its National Identification systems for cross-border verification of Ghanaians. It identifies what the current situation is, how it is affecting cross-border data flow, the opportunities that exist to deepen the borderless Africa agenda and what the government can do to address the challenges identified from the research. The research assesses conditions that offer the base for knowledge application and contribute to the literature for future research.

2.0 Research Methodology

The research adopts a mixed methodological approach where both qualitative and quantitative data are collected using surveys and interviews. The primary data collected is analysed to identify emerging themes and statistical observations to draw inferences. The first sample population consists of 100 data subjects, including both Ghanaians and non-Ghanaians living in Ghana. The second sample population consists of data controller and Personal Identification issuers, such as the National Identification Authority, Drivers' and Vehicle Licensing Authority, National Health Insurance Authority, and Social Security and National Insurance Trust. The data collection was done in two phases. The first phase has the survey targeted at 100 Ghanaian and non-Ghanaian citizens (data subjects) registered with any of the 5 Personal Identifier issuing institutions. The second phase involves interviews of designated officials of the 5 Personal Identification issuing institutions (data controllers) in Ghana.

The survey was administered online, while the interview was conducted in person. The data sets from the survey are analysed using descriptive statistics. Information gathered from the interviews and documentary reviews was analysed using narratives and thematic analysis. This was used to complement and triangulate the survey analysis to provide an in-depth understanding of the issues.

The survey gathered responses from 104 participants (data subjects) using a simple random sampling method. Additionally, 5 Personal Identification issuers (data controllers) were purposively selected for interviews. The Personal Identification issuing institutions in Ghana are the National Identification Authority (NIA), Electoral Commission (EC), Driver and Vehicle Licensing Authority (DVLA), National Health Insurance Authority (NHIA), and the Passport Office (PO). The foundational identification system is operated, maintained, and managed by the National Identification Authority. The functional identification systems include the National Health Insurance Authority, Driver's and Vehicle Licensing Authority, Social Security and National Insurance Trust, Passport Office, and the Electoral Commission.

3.0 Literary Context

This section discusses the local and continental level context within which the study was conducted, referencing existing literature on the topic to provide a better understanding of the underlying concepts.

3.1 Ghana's Identification Systems within the African Context

An Identification system is a technology adapted and implemented by governments to ensure that citizens' and non-citizens' vital biographic and biometric data are collected and stored on computer systems for analytical, planning, and development purposes.⁶ This system can be classified into foundational or functional. Foundational identification systems are designed to provide identification to the public to enable access to a wider range of services. Such systems include civil registers, national identification systems, and population registers. On the other hand, functional identification systems are created to provide identification, authentication, and authorisation for specific purposes, such as voting, driving, taxing, social security, and traveling.⁷ In Ghana, the National Identification Authority is the custodian of the foundational identification system and issuer of the national identity card (also known as the ECOWAS card) that serves as proof of legal identity. The institution is responsible for implementing and managing the national identification system in line with the National Identity Register Regulation, 2012. As of 2023, 16.3 million people have been issued with national identity cards.⁸ The Passport Office, Driver and Vehicle License Authority, National Health Insurance Authority, the Electoral Commission, and the Social Security and National Insurance Trust are custodians of functional identification systems and issue cards for specified use cases. The identity cards issued by these agencies are not considered proof of legal identity beyond the purpose for which they are issued. The national identity cards issued by the National Identification Authority have improved people's access to services, particularly banking services.⁹

Nonetheless, Ghana's foundational and functional identification systems remain fragmented, resulting in citizen fatigue and competing data infrastructure.¹⁰ An integrated system could assure users that all system elements function.¹¹ The lack of coordination in integrating information systems between government agencies that issue personal identification numbers causes replication and redundancy of population information data.¹²

6. Ebenezer A., et al (2019). *Formalizing Ghana's Economy through an Implementation of National Identification System: Issues and Perspectives*. Retrieved from https://www.internationaljournalcorner.com/index.php/ijird_ojs/article/view/146084/102529

7. Types of ID systems | Identification for Development. (n.d.). ID4D. Retrieved from <https://id4d.worldbank.org/guide/types-id-systems>

8. Awal, M. (2023). Over 800000 Ghana Cards uncollected. *The Business & Financial Times*. Retrieved from <https://thebftonline.com/2023/07/06/over-800000-ghana-cards-uncollected/>

9. Adjei Nkoju, E. C. (2021). Fintech tools, Ghana Card will improve SME access to Banking. *The Business and Financial Times*. Retrieved from <https://thebftonline.com/2021/09/16/fintech-tools-ghana-card-will-improve-sme-access-to-banking-umb-ceo/>

10. Thiel, A. (2020). Biometric identification technologies and the Ghanaian 'data revolution.' *The Journal of Modern African Studies*. Retrieved from <https://doi.org/10.1017/S0022278X19000600>

11. Gay, J. E. (2008). *Integration and Interoperability: An Analysis to Identify the Attributes for System of Systems*. DTIC. Retrieved from <https://apps.dtic.mil/sti/tr/pdf/ADA488988.pdf>

12. Zaini, R., Hidayat, F., & Rahardjo, B. (2020). *An Integration of National Identity towards Single Identity Number with Blockchain*. EUDL. Retrieved from <https://eudl.eu/doi/10.4108/eai.12-10-2019.2296532>

It also creates an absence of a central database. A recent study by Duodu (2021) revealed that the lack of a central database poses a major challenge to implementing the national identification projects in Ghana.¹³ An official of Ghana's foundational identification body in Thiel (2020) highlighted the need for a central database: "We need a central biometric system because scattered systems are easy to abuse. We are looking at it from a holistic national view. The same personal identification number will be used in all our registers. Being able to cross-reference will be the system's biggest value." The question of a central database is answered by integration and interoperability as it facilitates the exchange of data between different identification management systems for a specific objective.

This process of exchange may face technical difficulty due to the presence of varying protocols, standards, formats, and other similar factors.¹⁴ The complex interactions between regulations at different levels, and the fragmentation of systems can have implications on individuals and businesses, both intended and unintended.¹⁵ Although Ghana has the eGovernment Interoperability Framework to enhance open data systems and address data silos that exist between government agencies, the implementation has been ineffectual.¹⁶ It appears this situation also prevails in most parts of Africa. A World Bank study on the state of identification systems in Africa indicates that the integration and interoperability of identity systems are low in most parts of Africa. According to the study, there is a lack of unified identity management, as various agencies oversee different foundational identity systems, such as civil registration and identification. Additionally, multiple functional identity registers are maintained independently by various ministries, further contributing to the fragmentation of identity management.¹⁷

¹³. Duodu, E (2021). *Implementation of A National Identification System in Ghana : Lessons From The Indian Aadhaar System*.

Afribary. Retrieved from <https://afribary.com/works/implementation-of-a-national-identification-system-in-ghana-lessons-from-the-indian-aadhaar-system>

¹⁴. 6 Backhouse, J., Halperin, R. (2009). *Approaching Interoperability for Identity Management Systems*. In: Rannenbergh, K., Royer, D., Deuker, A. (eds) *The Future of Identity in the Information Society*. Springer, Berlin, Heidelberg. Retrieved from https://doi.org/10.1007/978-3-642-01820-6_

¹⁵. Fay, R. (2022). *Global Governance of Data and Digital Technologies: A Framework for Peaceful Cooperation*. Centre for International Governance Innovation. Retrieved from <https://www.cigionline.org/articles/global-governance-of-data-and-digital-technologies-a-framework-for-peaceful-cooperation/>

¹⁶ Ghana eGovernment Interoperability Framework (eGIF) Version 2.0. (n.d.). National Information Technology Agency. Retrieved from https://nita.gov.gh/theevooc/2022/02/Ghana-eGovernment-Interoperability-Framework-eGIF-Version2.0_v.Draft-for-Stakeholder-Review-1.pdf

¹⁶. Ghana eGovernment Interoperability Framework (eGIF) Version 2.0. (n.d.). National Information Technology Agency. Retrieved from https://nita.gov.gh/theevooc/2022/02/Ghana-eGovernment-Interoperability-Framework-eGIF-Version-2.0_v.Draft-for-Stakeholder-Review-1.pdf

¹⁷. Clark, J. (2017). *The State of Identification Systems in Africa*. Open Knowledge Worldbank. Retrieved from <https://openknowledge.worldbank.org/server/api/core/bitstreams/0406dff5-1fba-529b-b42a-b3cb3f87373b/content>

3.2 Data Governance and its related conceptual frameworks

Data governance can streamline operational efficiency and contribute towards the integration and interoperability of identification data systems. Through data governance, an organisation-wide programme can be created for assigning decision-related rights and duties to enable the adequate handling of data as an organisational asset.¹⁸ The goals of data governance include the removal of data silos by harmonising the data systems through a collaborative process, with stakeholders from the various participating units and to ensure that data is used properly, both to avoid introducing data errors into systems and to block potential misuse of personal and sensitive data of data subjects.¹⁹ This involves a systematic framework for classifying, organising, and communicating complex activities related to organisational data, and captures all aspects of the data pyramid, including operational, strategic, and executive levels. Guidelines and policies are also established for various stages of data, from creation to validation, as well as rules for ensuring data security and access at all levels. Data governance frameworks have several use cases, including master data management, data Integration and data warehousing, data migration and system consolidation, and regulatory compliance.²⁰

The DGI framework (Figure 1) is a notable conceptual data governance framework within which these goals can be achieved. A robust data governance framework is built upon several critical pillars, including technology, economy, security, law, and human rights.²¹ These key features ensure that data is managed efficiently, securely, and ethically, while also adhering to relevant legal and regulatory requirements. To maintain and manage a single authoritative data source, data custodians utilise data governance tools for master data quality management when implementing data governance frameworks.²²

¹⁸. Otto, Boris (2011) *Organizing Data Governance: Findings from the Telecommunications Industry and Consequences for Large Service Providers*. *Communications of the Association for Information Systems*: Vol. 29 , Article 3. Retrieved from, :
<https://aisel.aisnet.org/cais/vol29/iss1/3>

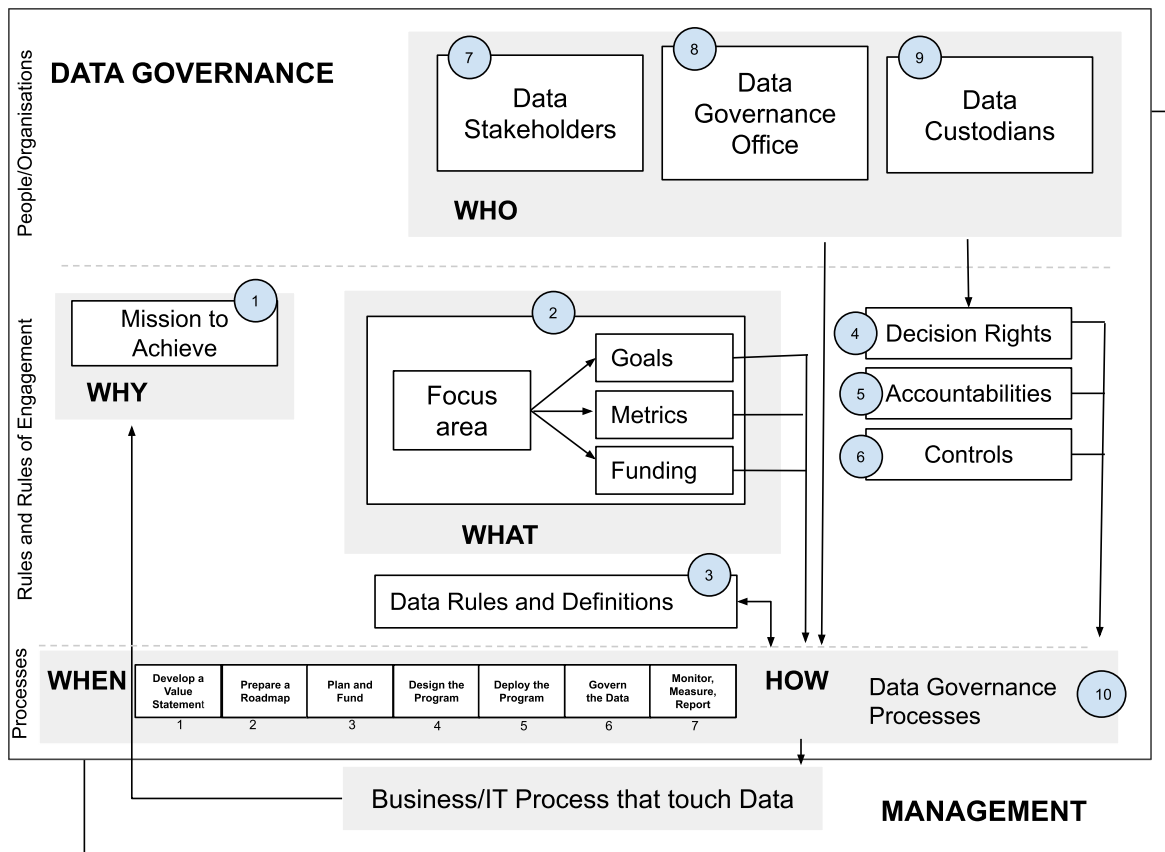
¹⁹. Stedman, C. (n.d.). *What is Data Governance and Why Does It Matter?* TechTarget. Retrieved from
<https://www.techtarget.com/searchdatamanagement/definition/data-governance>

²⁰. *What Are the 4 Data Governance Pillars?* (2023, July 11). DataGrail. Retrieved from <https://www.datagrail.io/blog/data-privacy/data-governance-pillars/>

²¹. *Strengthening Data Governance in Africa*. (n.d.). CSEA AFRICA. Retrieved from <https://cseaafrica.org/wp-content/uploads/2021/08/Strengthening-Regional-Data-Governance-in-Africa-1.pdf>

²². Arnaud, A. (2022). *Understanding What is Data Governance and Why is it Important*. To-Increase. Retrieved from <https://www.to-increase.com/master-data-management/blog/data-governance-definition-importance>

Figure 1: Data Governance Framework



Source: Data Governance Institute

Organisations implement data governance frameworks by following data governance operating models. These models represent how an organisation plans to leverage data to create business value. Through this, the capabilities required for planning, building, and running data products and data services are revealed.²³

The DAMA framework defines three types of data governance operating models which are centralised, federated, and hybrid. The specific data governance model that an organisation adopts depends on various factors, mainly the size of the organisation, the complexity of its systems, the industry it operates, and the regulatory environment.²⁴ The centralised model necessitates the creation of a single entity to take on absolute responsibility for all aspects of data governance, which includes control, coordination, and decision-making. The federated model embraces decentralisation by sharing the responsibilities for data governance among the various units, thus allowing for localisation but with oversight and coordination from a central entity. The hybrid model combines elements of both centralised and federated models allowing certain aspects of data governance to be centralised, while others are delegated to business units.²⁵ The approach to defining a data operating model takes into account the aspects of control, management, and existing capabilities in business units, but also the socio-cultural aspects of the organisation.²⁶

²³. Operating Model for Data Governance and Management. (n.d.). Business Technology Standard. Retrieved from https://www.managebt.org/content/uploads/BT_STD_Data_Governance_v1.pdf

²⁴. Data Governance Explained. (2023, March 2). AltexSoft. Retrieved from <https://www.altexsoft.com/blog/datagovernance/>

²⁵. Abousamak, K. (2023, July 26). Data Governance: Operating Models and Key Components. LinkedIn. Retrieved from, <https://www.linkedin.com/pulse/data-governance-operating-models-key-components-abousamak-pmp-cdmp/>

²⁶. Addagada, T. (2015, December). Choosing the Right Model of Operational Data Governance Initiatives. Retrieved from, https://www.researchgate.net/publication/368653457_Choosing_the_Right_Model_of_Operational_Data_Governance_Initiatives

3.3 Efforts Towards cross-border data flow across Africa

Achieving an integrated and interoperable national identification system can be the starting point for facilitating cross-border services in an increasingly digital global market. Estimates show that crossborder data flows added \$2.8 trillion to the global Gross Domestic Product and is expected to grow to \$11 trillion by 2025.²⁷ Although cross-border data sharing can bring significant advantages to businesses, individuals, and entire nations, many African countries at various stages of development have created regulatory and structural obstacles to such data flows. These barriers can take the form of data residency requirements that restrict data within a country's borders, a practice commonly referred to as "data localisation", which is rising as countries enact barriers to data flows to address privacy, cybersecurity, and economic mercantilism.²⁸

To harness digital technologies and innovations that promote integration and bridge the digital divide, the African Union adopted the Digital Transformation Strategy for the period 2020 to 2030. The strategy, among other objectives, seeks to promote open standards and interoperability for crossborder trust framework, personal data protection, and privacy in light of the creation of the African Free Trade Area.²⁹ Because digital transformation comes with the production of data, the African Union also developed and approved the AU Data Policy Framework to set out a common vision, principles, strategic priorities, and key recommendations to guide member states in developing national data systems and capabilities to effectively use and derive value from data.³⁰ Studies suggest that Africa needs more integrated and interoperable identification systems. This is crucial for enabling regional mobility and economic integration. Standards are important in ensuring that the building blocks of identity systems are interoperable and testable and can meet desired performance targets. Mittal (2022) outlines the technical standards for implementing and maintaining digital identification systems.³¹ The AU Interoperability Framework for Digital ID provides a common standard at the continental level to represent digitally, the proofs of identity issued by trusted sources from member states and to ensure interoperability throughout the continent. The Framework defines common requirements, minimum standards, governance mechanisms, and alignment among legal frameworks.³² The goal is to make it possible for every African citizen to access public and private services conveniently and securely, regardless of their location. ECOWAS, a block within the African Union, introduced the ECOWAS National Biometric Identity Card (ENBIC) to enable interoperability and mutual recognition of identity credentials among Member States. West Africa is a highly mobile region with a significant portion of its cross-border movements being intra-regional migration.³³

27. Data Flows - ICC. (n.d.). International Chamber of Commerce. Retrieved from <https://iccwbo.org/global-insights/digital-economy/data-flows/>

28. Cory, N. (n.d.). Cross-Border Data Flows: Where Are the Barriers, and What Do They Cost? Itif. Retrieved from <https://www2.itif.org/2017-cross-border-data-flows.pdf>

29. African Union. (n.d.). African Union. Retrieved from <https://au.int/sites/default/files/documents/38507-doc-dtsenglish.pdf>

30. AU Data Policy Framework. (n.d.). African Union. Retrieved from <https://au.int/sites/default/files/documents/42078-docAU-DATA-POLICY-FRAMEWORK-ENG1.pdf>

31. Mittal, A. (2022). Catalog of Technical Standards for Digital Identification Systems. Work Bank. Retrieved from <https://documents1.worldbank.org/curated/en/707151536126464867/pdf/Catalog-of-Technical-Standards-for-DigitalIdentification-Systems.pdf>

32. AU Interoperability Framework for Digital ID. (n.d.). African Union. Retrieved from https://au.int/sites/default/files/documents/43393-doc-AU_Interoperability_framework_for_D_ID_English.pdf

33. West Africa Moves Towards Biometric Identity Cards. (2015). IOM. Retrieved from <https://www.iom.int/news/westafrica-moves-towards-biometric-identity-cards>

In Ghana, under the Office of the President, the National Identification Authority is mandated to register, print, and issue the ECOWAS identity cards (also known as the Ghana Card) to citizens and non-citizens.³⁴ The card is equipped with both contact and contactless smartcard interfaces and features a 148K memory chip. It is printed with strict adherence to the global standards set by the International Civil Aviation Organisation (ICAO) and the National Institute of Standards and Technology to ensure its security and consistency. Of the 15 member states, only Senegal, Guinea-Bissau, Ghana, Benin, The Gambia, and Sierra Leone have ECOWAS National Biometric Identity Card.³⁵ This serves as the foundational identification system of these member states.

At scale, the scope of the ECOWAS National Biometric Identity Card (ENBIC) could be expanded to work like the European Union's electronic Identification, Authentication and Trust Services (eIDAS) regulation, which promotes interoperability and the mutual recognition of the electronic identification schemes of member European Union states. The regulation provides legal backing for the trust services offered by organisations that meet the specified requirements. The eIDAS Network comprises interconnected eIDAS-Nodes, one for each participating country. These nodes enable cross-border authentication requests and responses. By connecting to the eIDAS node, Service Providers, including public administrations and private sector organisations, can make their services accessible across borders and benefit from the legal recognition provided by eIDAS.³⁶ The benefit accruing from this regulation makes it possible for Europeans to open bank accounts, establish businesses, and sign contracts electronically for conducting trade in European Union member-state countries without requiring any physical presence. The eIDAS solutions also help businesses carry out cross-border transactions.³⁷

There are three broad potential architectures to facilitate mutual recognition of national identity cards while maintaining national sovereignty. They include (1) *web-based*: online web-based authentication using the federation protocol (SAML or Open ID connect). Similar architecture is used under eIDAS; (2) *API-based*: online authentication using an API approach. Similar architecture is used among some Latin American countries; and (3) *Public-Private Key-based*: offline and online authentication verifying the private key on a credential against a public key directory. Similar architecture is used for the ICAO Public Key Directory of electronic passports. The diagram (Fig. 2) of the web-based architecture adopted by the EU for the eIDAS is presented below.

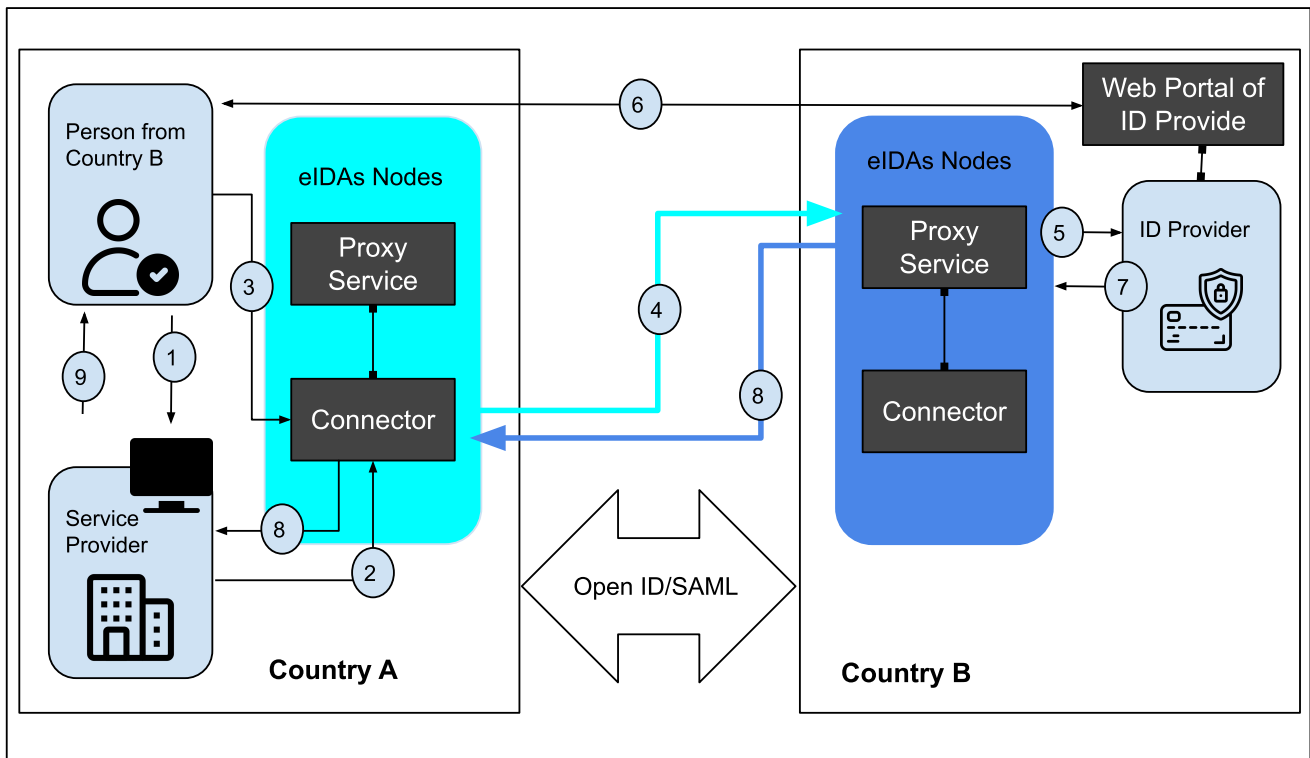
34. Parliamentary Committee Report on the Status of Registration, Printing, and Issuance of Ghanacard and its Challenges. (n.d). Retrieved from <https://ir.parliament.gh/bitstream/handle/123456789/2706/ECOWAS%20CARD.pdf?sequence=1&isAllowed=y>

35. Ponle, T. (2023). Only 6 member states have ECOWAS biometric card – Commission. Citizenship Rights in Africa Initiative. Retrieved from <https://citizenshiprightsafrica.org/only-6-member-states-have-ecowas-biometric-card-commission/>

36. Mutual recognition of IDs across borders | Identification for Development. (n.d.). ID4D. Retrieved from <https://id4d.worldbank.org/guide/mutual-recognition-ids-across-borders-0>

37. Discover eIDAS | Shaping Europe's digital future. (2023). Shaping Europe's digital future. Retrieved from <https://digitalstrategy.ec.europa.eu/en/policies/discover-eidas>

Figure 2: Web-based mutual recognition architecture



Source: World Bank

Authentication Flow

1. A person from Country B accesses a service on a browser via the website of a service provider in Country A.
2. The Service provider sends the request to the Connector in Country A
3. The Connector in Country A asks the person from Country B for their country of origin, if not already provided.
4. The Connectors in Country A forward the request to the Proxy Service of Country B.
5. The Proxy Service B sends the request to the Identity Provider in Country B for authentication (the person in Country A is redirected to the login page of the identity provider in Country B).
6. The person logs into the web portal of the Identity provider of Country B in Country A.
7. Once authenticated, the Identity provider in Country B sends a response that is returned to Proxy Service B
8. Proxy Service B sends a SAML Assertion to the requesting Connector A, which forwards this response to the Service Provider located in Country A (the person's browser is redirected back to the Service Provider's website).
9. The Service Provider grants access to the person.

According to the World Bank, the eIDAS architecture works best with the Federation protocol implementation. That is Security Assertion Markup Language (SAML) or OpenID Connect Server (eIDAS-node). There must be a Web portal for user authentication to be provided by the identity provider. With most African countries having low digital literacy rates, it is recommended that people should be educated to authenticate their identity using passwords, One-Time Passwords (OTP), Personal Identifiable Numbers (PIN), or Fast Identity Online (FIDO) authenticators.

3.4 Standardising Data Protection and Privacy Regulations

Regulations are a key element in making cross-border identification authentication possible. Yang (2021) highlighted two regulatory approaches for cross-border data flow. The EU supervisory approach: a dual supervisory approach based on geography stipulates that state parties cannot restrict cross-border data flows solely on the grounds of protecting privacy. It also applies the “sufficient protection” principle that requires third countries to have similar and adequate protection when data is transferred outside a member state in line with the GDPR. The Regulatory approach of the US: the principle of accountability as core establishes privacy protection standards in the American industry and allows the default data controller or processor to comply with relevant regulations in their business activities and take responsibility for data security legally and reasonably. Although the US promotes cross-border data flow, it practices data protectionism for key technologies and security.³⁸

In Ghana, the law establishing the National Register, Act 750, as amended by Act 950 and operationalised by the Legislative Instrument, L. I 2111 makes provision for the access to information to user agencies. There are currently no cross-border data flow restrictions that are applicable, although the sale, purchase, knowing, or reckless disclosure of personal data is prohibited under the Data Protection Act.³⁹ Creating internationally compatible data protection regimes builds trust online and predictability for all stakeholders in the information economy.⁴⁰ Standardisation of data protection regulation can provide a solution for responsible cross-border data flows.⁴¹ Towards this end, the ECOWAS Supplementary Act A/SA.1/01/10 on data protection seeks to establish a legal framework of protection for the privacy of data and use of data without prejudice to the general interest of member states. The agreement requires member states to establish a data protection authority. To date, only 7 countries (including Ghana) have enacted legislation in compliance with the agreement.⁴² The AU Data Policy also provides a non-binding framework for the handling of data. A CIPESA brief published in 2022 noted limited domestic funding to data infrastructure and data ecosystems across Africa remains a challenge in the implementation of this framework. The author argued that financing for data needs to be held with the same importance as financing for roads and hospitals, as the former is also a public good, especially in the current information age.⁴³

38. Yang, X. (2021). *Regulatory Approaches of Cross-border Data Flow in the Big Data Era: China's Choice*. *Journal of Physics: Conference Series*. <https://iopscience.iop.org/article/10.1088/1742-6596/1848/1/012026/pdf>

39. Africa Guide to Data Protection. (n.d.). Bowmans. Retrieved from https://bowmanslaw.com/wpcontent/uploads/2022/06/Data-Protection_01.06.2022.pdf

40. Biersteker, N. (n.d.). *Data protection regulations and international data flows: Implications for trade and development*. unctad. Retrieved from https://unctad.org/system/files/official-document/dtlstict2016d1_en.pdf

41. Sandor, L. (2023, May 31). *Can Standardisation in Policies be a Solution for Responsible Cross-Border Data Flow for Health-Related Data?* *Datasphere Initiative*. Retrieved from <https://www.thedatasphere.org/news/can-standardization-in-policies-be-a-solution-for-responsible-cross-border-data-flow-for-health-related-data/>

42. *Review of e-commerce legislation harmonization in the Economic Community of West African States - ECOWAS*. (n.d.). unctad. Retrieved from https://unctad.org/system/files/official-document/dtlstict2015d2_en.pdf

43. Sabiti, B. (2022, October). *Five Takeaways From the 2022 African Union Data Policy Framework Brief*. CIPESA. Retrieved from https://cipesa.org/wpcontent/files/briefs/Five_Takeaways_From_the_2022_African_Union_Data_Policy_Framework_Brief.pdf

4.0 Results and Findings

This section presents results and findings from the analysis of data collected on the various population segments identified for this study.

4.1 Results

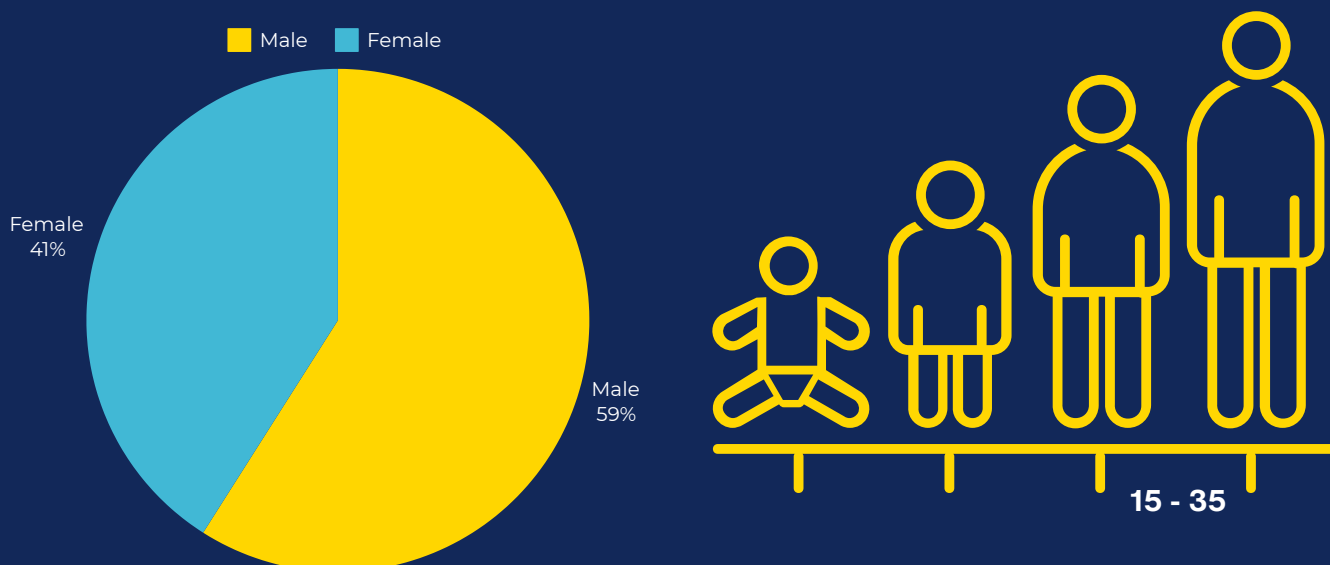
The results of the survey of data subjects and the interview of the data custodians sampled for this study are presented in this section.

4.1.1 Data Subjects

This segment presents an analysis of responses collected from Data Subjects. These are individuals, both Ghanaians and non-Ghanaians living in Ghana, who have one or more identification cards issued by the Data Controllers - Personal Identification issuing bodies with the mandate to maintain functional or foundational identification systems.

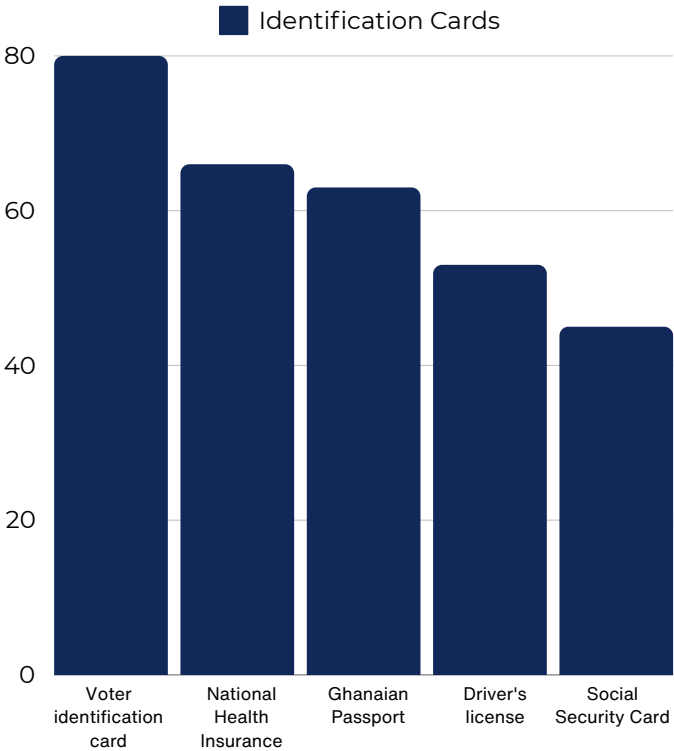
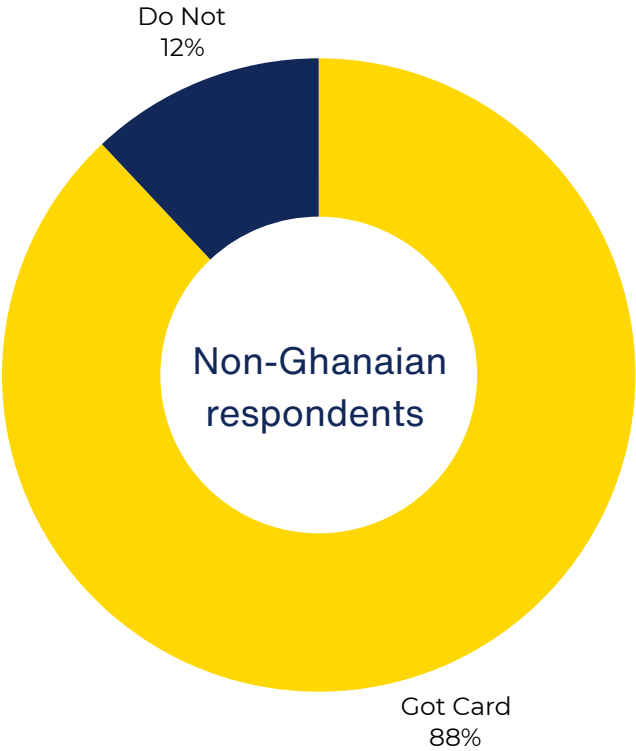
4.1.1.1 Demography

The survey was conducted online and received 104 responses from both male and female participants, who were either Ghanaians or non-Ghanaians residing in Ghana. The non-Ghanaians included citizens of Kenya, Nigeria, and South Africa. The male respondents constituted 59% of the total, while females accounted for 41%. It can not be verified that this was influenced by the digital gender gap. The largest portion of the respondents, representing 48%, fell within the 15 to 35 age range. It's worth noting that the respondents are highly educated, with 93% holding tertiary-level qualifications.



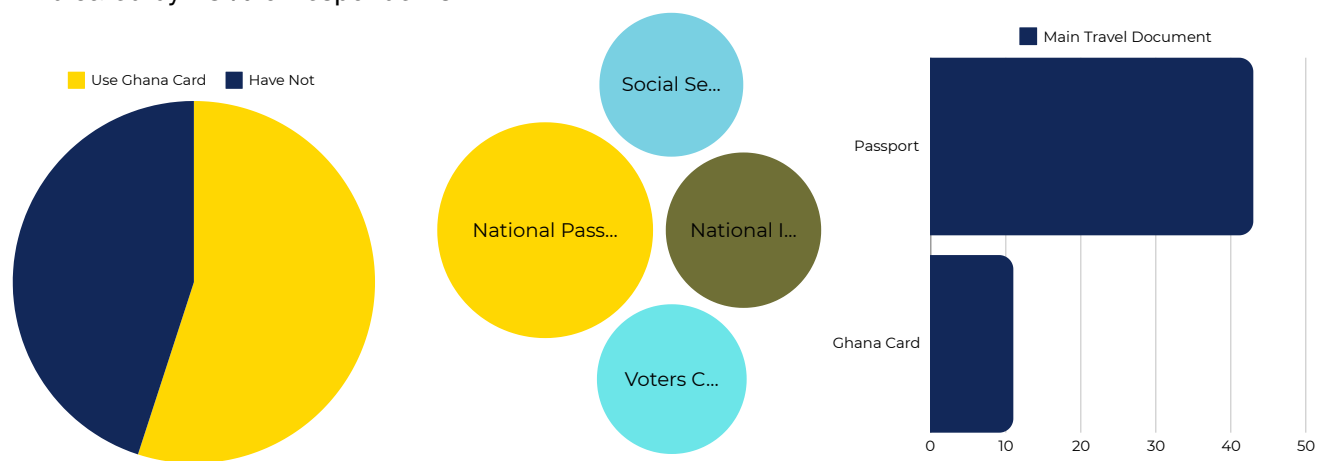
4.1.1.2 Identity Card Ownership

The vast majority of the non-Ghanaian respondents, accounting for 88%, possess the national identity card of their country of origin, while the remaining 12% do not. However, 98% of all respondents, including both Ghanaians and non-Ghanaians, have the Ghana Card, which is the primary identification document of the Republic of Ghana. Ghanaians obtained the Ghana Card at no cost, whereas non-Ghanaians paid an average of US\$ 62. The majority of the respondents, representing 32%, acquired their cards in 2019. The fee has been increased to US\$ 75 for non-Ghanaians from the rest of Africa in 2024. Non-Ghanaian citizens receive the Ghana Card instantly, while Ghanaian citizens typically receive theirs a few days after registration. The respondents hold multiple identification cards. Apart from the Ghana Card, 80% have a Voter identification card, and 66% possess a National Health Insurance Card. Similarly, 63% hold a Ghanaian passport, while 53% and 45% have a Driver's license and a Social Security card, respectively. It was noted that none of the non-Ghanaian respondents held a Voter identification card. The base document provided used to acquire the other identification cards was the Ghana Card, according to 63% of the respondents. In the absence of a Ghana Card, respondents provided other base documents which include Birth certificate (Ghanaian), Passport (Ghanaian or non-Ghanaian), and Driver's license (foreign).



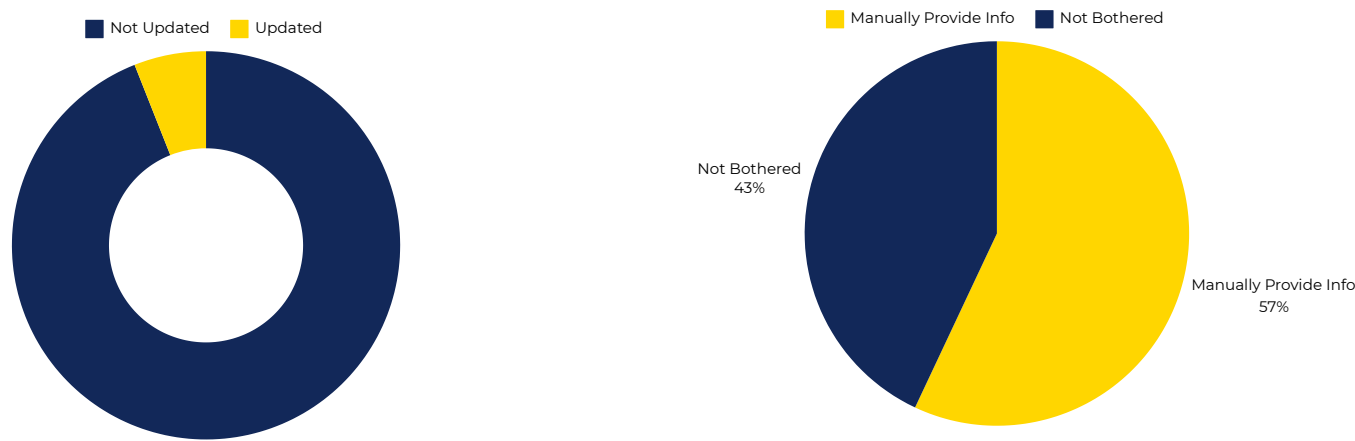
4.1.1.3 Using the Ghana Card in Place of Other Identity Cards to Access Related Services

Respondents use their foundational identity cards to verify their identity when utilising services provided by the issuers of the functional identity cards. The data indicates that 55% of the respondents have used their Ghana cards in lieu of other functional identity cards in the past, while 45% have not. They primarily used their Ghana Card in place of the National Passport (27%), National Insurance Card (14%), Social Security Card (13%), and Voter's Card (12%). In the scenarios outlined, the respondents utilised their Ghana Cards mainly for domestic and regional travel, as well as to access financial services such as pensions and banking. Respondents have never used the Ghana Card to vote in an election. Although some respondents (11%) have used their Ghana card for traveling, the main travel document remains the National passport, as indicated by 43% of respondents.



4.1.1.4 Interoperable Information Updates Across Multiple Identify Cards

On average, each respondent has 4 identity cards, including the Ghana card. These cards will need to be updated as the cardholder's circumstances change over time. At the time of the survey, the majority of respondents, representing 94%, had not updated the information on their Ghana cards, while only 6% had done so. Among those who had updated their cards, 57% had to manually provide the same information at different times to the issuers of their other cards. The remaining 43% had not bothered to update their other identity cards with the new information on their Ghana cards. It is worth noting that respondents make updates to the information on their cards individually and usually at a cost that includes time and finances.



4.1.1.5 Cross-border Usage of the Ghana Card

As has been established, the majority of respondents, representing 98%, possess Ghana cards which they use for a variety of purposes, including accessing public services, travelling, financial services, etc. The data shows that only 8% of the respondents with Ghana cards have ever utilised their cards abroad to access services, while 7% had tried, but their card was not accepted. The majority of respondents, representing 85%, have never tried to use their Ghana card in another country. Those who utilised their Ghana cards in another country have done so mostly in sub-regional countries to access immigration services only. Similarly, non-Ghanaians in Ghana are also not able to utilise the identity cards from their country of origin in Ghana, except for border crossing purposes at the port of entry, as indicated by 14% of respondents.

4.1.1.6 Data Privacy Concerns

Respondents noted they have concerns when it comes to the privacy and protection of their data. This is according to 38% of the respondents. Their concerns were directed at the National Identification Authority and the Electoral Commission, representing 31% and 17%, respectively. One respondent noted: “My main concern is whether their database is safe not to leak personal information to hackers or people getting access to citizens' information when they don't have the right to get it”. Another recounted: “The National Communication Authority directed that all phone numbers must be registered with the Ghana card. Later, a short code was published to enable people to check their registration status. I've heard cases of people's Ghana cards being used by unknown persons to register their phone numbers. With this, you don't feel protected or have privacy with your information.” There were also concerns about solicitation, as outlined by a respondent this statement: “I get unnecessary messages from advertising firms, and I believe they got my details thanks to the organisations mentioned above”

4.1.1.7 Digital Savvy

Respondents have high literacy with most having tertiary level education. It appears this showed in their digital literacy skills, with 37% and 34% of respondents rating themselves on level 4 and 5 competencies, respectively. The appreciation of information technology and electronic device usage is high among respondents. They understand the benefits and risks associated with the usage of the Ghana card. Implementing public education is made easy since most data subjects are digital savvy.

4.1.2 Data Controllers

This segment presents an analysis of the interview conducted to collect information from the Data Controller. These are Personal Identification issuing institutions in Ghana with the mandate to maintain functional or foundational identification systems and issue identity cards. Using a purposive sampling technique, five participants, also known as Key Informants (KIs), were drawn from the Personal Identifier issuing bodies. These are the National Identification Authority (NIA), Electoral Commission (EC), Driver and Vehicle Licensing Authority (DVLA), National Health Insurance Authority (NHIA) and the Passport Office (PO). For ease of identification and confidentiality, Key informants were marked as shown in the table below:

Table 1: Key informants

S/N	Key Informant	Mark	Gender Distribution
1	National Identification Authority (NIA)	KI A	Male
2	Electoral Commission (EC)	KI B	Female
3	Driver and Vehicle Licensing Authority (DVLA)	KI C	Female
4	National Health Insurance Authority (NHIA)	KI D	Male
5	Passport Office	KI E	Male

4.1.2.1 Briefs

A summary of the agencies interviewed for this study is presented below. The information captures Acts establishing the personal identifier issuing institutions and their mandate.

Agency	ID System Classification	Description
National Identification Authority (NIA)	Foundational	Established in 2003 under the Office of the President with the mandate to issue national ID cards and manage the National Identification System (NIS). The NIA Act, 2006 (Act 707) was passed to give it the necessary legal premises on which to operate. The National Identity Register Act, 2008 (Act 750) was also passed to give authorisation for the collection of personal and biometric data and to ensure the protection of the privacy and personal information of enrollees. ⁴⁴
Electoral Commission (EC)	Functional	Established by the Electoral Commission Act (Act 451) of 1993 with the mandate to manage the conduct of all public elections and Referenda, and to handle all matters directly related to the conduct of elections in the country. It is one of the governance institutions provided under the 1992 constitution. ⁴⁵
Driver and Vehicle Licensing Authority	Functional	Established in 1999 by an Act of Parliament (Act 569), 1999, to provide a regulated framework for an enhanced and more effective administration of drivers and vehicles. ⁴⁶
National Health Insurance Authority (NHIA)	Functional	Established under the National Health Insurance Act 2003, Act 650 replaced Act 852 to attain universal health insurance coverage concerning Ghanaians and nonGhanaians resident in Ghana. ⁴⁷

⁴⁴. About Us. (n.d.). National Identification Authority. Retrived from <https://nia.gov.gh/our-history/>

⁴⁵. About Us - Electoral Commission. (n.d.). Electoral Commission of Ghana. Retrieved from <https://ec.gov.gh/about-us/>

⁴⁶. About Us. (n.d.). DVLA. Retrieved from <https://dvla.gov.gh/about.php>

⁴⁷. NHIA - Accra. (n.d.). NHIS. Retrieved from <https://www.nhis.gov.gh/nhia>

Agency	ID System Classification	Description
Social Security and National Insurance Trust (SSNIT)	Functional	Established in 1972 under NRCD 127, replaced by the National Pensions Act, 2008 Act 766 to administer Tier 1 of the Three-Tiered Pension Scheme. It caters to both Ghanaians and non-Ghanaians. ⁴⁸
National Passport Office	Functional	A bureau of the Ministry is responsible for the issue of passports and other travel documents of Ghana to Ghanaian Citizens in accordance with the Passports and Travel Certificate Decree 1967, (NLCD 155). ⁴⁹

4.1.2.2 Integration of Ghana's Identification Systems

Preliminary evidence gathered from the interview has shown evidence of data integration of Ghana's identification system. It appears a process is ongoing to consolidate key data sets from the functional identity card issuers into the database of the National Identification Authority so that the Ghana card can function as a universal card. This enables individuals with Ghana cards to access services associated with the functional identity card issuers without holding the cards originally issued by these institutions. Speaking to the issue of data integration of Ghana's Personal Identifier issuing institutions, the Key Informant for the National Health Insurance Authority stated:

Text 1: Key Informant, National Health Insurance Authority

"...We keep improving on our system. Previously, we used not to take the biometric details of beneficiaries and our services were district-based. A beneficiary from one district could use the card to access services in another district. But now, the law has been amended to allow us to take the biometric details of beneficiaries when they are registering. This has also allowed our data to be integrated nationally. Now, beneficiaries can access services anywhere once they hold the card. On integration with other identity issuers, holders of Ghana cards who are registered with the National Health Insurance Scheme can access our services using the Ghana Card. Additionally, you can register for the NHIS online when you have the Ghana Card..."

The views separately expressed by the Key Informant from the National Identification Authority corroborate those earlier expressed by the Key Informant from the National Health Insurance Authority.

⁴⁸. About Us – SSNIT. (n.d.). SSNIT. Retrieved from <https://www.ssnit.org.gh/about-us/>

⁴⁹. Services – Ministry Of Foreign Affairs and Regional Integration. (n.d.). Ministry Of Foreign Affairs and Regional Integration. Retrieved from <https://mfa.gov.gh/index.php/services-2/>

Text 2: Key Informant, National Identification Authority

“As we move towards a single card agenda, the National Identification Authority is collaborating and supporting other identity issuing agencies to achieve complete data integration and to consolidate all the separate databases into a single repository. So, you’ll notice that holders of the Ghana card can now access other services using the Ghana card. For example, you can integrate data on your passport with your Ghana card, and you wouldn’t have to carry your license around when driving. Also, in some African countries, your Ghana Card is enough to serve as a passport for you...”

The views expressed by the Key Informants from the National Identification Authority and National Health Insurance Authority are supported by the survey results outlined in 4.1.3, which indicates how Ghana cardholders can utilise their card to access services related to the functional identity card issuers.

4.1.2.3 Interoperability of Ghana’s Identification Systems

Although the government of Ghana has developed and adopted the e-Governance Interoperability Framework to define standards and policies that would guide the selection of technology, channels, etc to enable broader interoperability among government agencies, the implementation has been ineffective. Evidence suggests there is no system interoperability among the various identity card issuers in Ghana. Speaking to the Key Informant from the National Identification Authority on interoperability, he revealed that even though the Authority has plans for interoperability with the databases of the functional identity card issuers, that has not been achieved yet. The lack of operational capacity, funding, and different data management systems among the personal identifier issuing institutions is hampering progress toward system interoperability.

Text 3: Key Informant, National Identification Authority

“The legislation establishing the National Identification Authority envisages and encourages interoperability due to the need to have a single card in the future. Because of the objective of gradually building a single database for all the issuers, there’s currently significant effort at realizing interoperability among the various issuers... I must admit that the process is slow and hampered by lack of operational capacity, the needed investment, and the regularisation of the data management systems to enable interoperability...”

This explains why survey respondents who have ever made updates to the information on their cards have had to do so separately to each of the cards at different times, as found in 4.1.4. Similarly, the statements from the Key Informant from the Electoral Commission corroborate that of the Key Informant from the National Identification Authority;

Text 4: Key Informant, Electoral Commission

“Our system is not interoperable with any other system, and this is partly due to privacy and data security concerns. This may happen in the future, but for now, I don’t know of any such arrangement to make our system accessible to any other institution in the name...”

4.1.2.4 Cross-border Data Transfer from Ghana's Identification Systems

The information gathered from the interviews of the personal identifier issuing institutions indicates that neither the foundational nor the functional identification systems are integrated into another international system. With the absence of system-level integration, interoperability is rarely achieved. This explains why survey respondents have never been able to utilise the Ghana cards in any other country for use cases other than regional travel and border crossing, as established in 4.1.5. The Ghana card is useful for regional travel because it is issued with strict adherence to the global standards set by the International Civil Aviation Organization (ICAO) and the National Institute of Standards and Technology to ensure its security and consistency. Speaking to the Key Informant from the National Identification Authority on the cross-border usage of the Ghana card, he confirmed it as a travel document but highlighted implementation delays as some of the challenges on the part of member countries that are hindering a broader use case.

Text 5: Key Informant, National Identification Authority

“The Ghana card can be used for international travel within the ECOWAS region when countries in the region begin to use e-gates at their borders. Its features are compliant with ICAO standards. Some citizens of Ghana have traveled across the borders of Ghana within the region and were admitted based on their Ghana card. However, not all countries are ready to accept the Ghana card at the moment and this is partly due to a lack of implementation of the appropriate systems to read the cards. Also, whereas in Ghana, the Card can be used for a variety of services including banking and telecommunications, these services are yet to be made accessible in member states for West African citizens who hold the ENBIC...”

4.2 Findings

Aside from the call for unified and unambiguous legal regimes for handling data across member states, the African Union Data Policy also proposes the standardisation of data governance practices across key government institutions to enable complementary levels of obligations. A component of the AU Data Policy Framework focuses on integration and interoperability. The study assessed Ghana's data governance and the integration and interoperability of its identification system within the conceptual framework of the AU Data Policy Framework. The following are the findings

1. Non-alignment with the African Union Data Framework

To address the fragmentation in the Government of Ghana's IT systems across ministries, departments, and agencies to achieve standardisation and interoperability, the government developed and adopted two documents - the Ghana Government Enterprise Architecture and the e-Government Interoperability Framework. Indications are that the Implementation of the first version of these documents did not yield the desired results mainly due to the lack of a centralised implementation body, non-compliance, lack of education, and insufficient funding. These documents have been reviewed to take cognisance of disruptive technologies. The implementation of the reviewed version is expected to achieve better outcomes than the first. The objectives of these documents do not align with the African Union Data Policy Framework. The Ghana Government Enterprise Architecture, based on 15 principles, mentions nothing about cross-border data access. The same applied to the e-Government Interoperability framework which is also based on 11 principles with no provision for crossborder data flow. Both documents appear to be tilted more towards data localisation, essentially focusing on Ghana's ministries, departments, and agencies. The AU Data Policy Framework aims to facilitate cross-border data access among member states. In making a case for an open system, the e-Government Interoperability framework recommends that ministries, departments, and agencies encourage openness when working with each other to establish digital services for the Government while considering their priorities and constraints. Furthermore, the ministries, departments, and agencies are advised to ensure technological neutrality and adaptability. This may be a good point to shape Ghana's data governance. Nonetheless, it seems that the implementation of the second version of the Ghana Government Enterprise Architecture and the e-Government Interoperability Framework may encounter challenges as none of the institutions responsible for issuing personal identifiers, interviewed as part of this study, was aware of the existence of these documents. Taking stock of existing policies on data shows that Ghana does not have a data governance framework in place to manage data from the various ministries, departments, and agencies. However, there is a data-sharing policy that establishes a governance framework for the sharing of government data, including the implementation of the National Open Data Initiative.

2. Existence of data integration

The component of system integration that currently exists among the Personal Identifier issuers in Ghana is data integration. Infrastructure, application, and process integration of the various identification issuers are non-existent. Data integration involves consolidating data from multiple sources into a single and unified view. Specific data that cardholders have with the functional identification issuers are collected by the foundational identification issuer from the individuals at the point of registering. The National Identification Authority acts as the central repository or data warehouse where the identification numbers that individuals have with the various functional identification issuers are captured. For example, at the point of registration with the National Identification Authority for a Ghana card, the data subject is asked to provide their driver's license, SSNIT, Passport, and NHIA identification number. Data integration should not be mistaken for system integration. The former focuses on consolidating data from multiple sources into a single storage for a unified perspective whereas the latter involves linking various systems to enable information access.

3. Non-existent of interoperability

Integration of the systems of Ghana's Personal Identifier issuing institutions would involve establishing connections and protocols that enable seamless data exchange and interaction between different software and hardware platforms of the identification issuers. This would have resulted in system interoperability, thus enabling two-way data exchange. The level of integration that currently exists does not facilitate the achievement of interoperability. This explains why the survey respondents have to update information on their cards separately with each of the Personal Identifier issuing bodies. An update of information on the Ghana card would not automatically reflect on the individual's information with the functional Personal Identification issuing identification issuers unless the individual goes and makes the changes there, too.

4. No cross-border data flow

The transfer of data across international borders necessitates a network of interconnected servers spanning different countries. This enables seamless movement of information between participating nations, ensuring that citizens located anywhere on both sides of the border have unfettered access to the data and services they require. The Ghana card is not accepted in any other country as a legal document to prove the identity of resident and nonresident Ghanaians to enable them access to services such as banking, telecommunication, ecommerce, etc. The information technology system of Ghana's foundational identification issuer is not integrated with any counterpart in any other country. The Ghana card can only be used for travels within ECOWAS because it is ICAO compliant and has the features of an epassport, which is suitable for travels where immigration stamps are not required. That is why some survey respondents were only able to use it for travel to neighbouring West African countries. This is also the case for non-Ghanaian residents in Ghana. They would have to acquire a non-citizen Ghana card to be able to access services in Ghana.

5. Data protection and privacy concerns

Survey respondents have concerns about the protection and privacy of their data. They have reported cases of unknown people using the details of others to acquire services such as online borrowing, SIM card registration, e-commerce, etc. There have also been incidences of unwarranted solicitations and consent issues. According to the National Identification, there are strict information-sharing guidelines in place for all agencies querying the database of the Authority through the Identity Verification System Platform. The Authority does not allow foreign entities to connect to this system.

5.0 Discussion

Over the years, Ghana's identification and digital system has gone through significant strides. Preliminary evidence gathered from the data collection and analysis has shown evidence of data integration of Ghana's identification system. A key informant from the National Identification Authority confirms this.

"As we progress toward a unified identification system, the NIA is working with other identity issuing agencies to achieve full integration and merge all separate databases into a single repository. As a result, holders of the Ghana card can now access various services using the same card. For example, you can link your passport to your Ghana card and won't need to carry your driver's license when driving. Additionally, in several African countries, the Ghana Card is sufficient as a passport."

Data integration within Ghana's identification system only consolidates data from the various functional identification issuers into a central repository managed by the National Identification Authority. This form of integration enables a unified view of individuals' data, facilitating the use of the Ghana card for accessing various services. However, this integration is limited to data and does not extend to infrastructure, applications, or processes. The consolidation of identification numbers from different functional identification issuers into a single database is a positive step toward achieving a comprehensive identification system. However, the absence of system integration means that data updates must be manually provided to each functional identification issuer, resulting in inefficiencies and potential data inconsistencies. This poses a significant inconvenience to the identity card holder, as mentioned by a respondent in the survey.

"It costs in terms of transportation and time. As a result, one has to decline every job opportunity for that day and hope the task is completed. The money that one could have earned that day will not be attainable."

Ghana's data governance structures, as embodied by the Ghana Government Enterprise Architecture and the e-Government Interoperability Framework, are not fully aligned with the African Union Data Policy Framework. While these documents aim to standardise and promote interoperability within Ghana's ministries, departments, and agencies, they do not adequately address cross-border data access. The lack of a centralised implementation body, insufficient funding, and limited awareness among key institutions further compound these challenges. Consequently, Ghana's current focus appears to lean toward data localisation, potentially hindering efforts toward regional data integration and exchange.

Although the foundational identification card issued by the National Identification Authority is ICAO compliant and can be used for regional travel within the ECOWAS region, it is not recognised as a legal document for accessing services such as banking, telecommunications, or e-commerce in other countries. This limitation underscores the lack of international system integration and highlights the need for Ghana to engage in regional and international frameworks that facilitate cross-border data exchanges for citizen verification.

Despite the National Identification Authority's assertion of strict information-sharing guidelines, the incidence of data breaches and unwarranted solicitations indicates gaps in the enforcement and effectiveness of these measures. These concerns underscore the need for robust data protection frameworks and stronger enforcement mechanisms to safeguard personal data against misuse and ensure privacy.

6.0 Recommendations

Based on the findings from the study, the following recommendations are made for the consideration of relevant stakeholders, including government, civil society, and academia.

6.1 Government

1. Draft a data governance framework

The Ghanaian government should consider drafting and adopting a national data governance framework that aligns with the African Union Data Policy to encourage integration, interoperability, and cross-border data flow. This should, among other data controllers, include the Personal Identifier issuing bodies. Access levels must be defined for internal and external data users while protecting the right to privacy for data subjects. The DGI Framework could be a good reference model. The data governance framework can complement the Ghana Government Enterprise Architecture, the e-Government Interoperability Framework, and data sharing policy.

2. Effective implementation of data governance framework

The Ghanaian government should consider drafting and adopting a national data governance framework that aligns with the African Union Data Policy to encourage integration, interoperability, and cross-border data flow. This should, among other data custodians, include the Personal Identifier issuing bodies. Access levels must be defined for internal and external data users while protecting the right to privacy for data subjects. The DGI Framework could be a good reference model. The data governance framework can complement the Ghana Government Enterprise Architecture, the e-Government Interoperability Framework, and data sharing policy.

3. Establish a bridge system for cross-border data flow

A national centralised bridge system connection to the National Identification Authority's Identity Verification System Platform should be established. This will allow foreign counterparts to integrate, under bilateral cross-border data flow arrangements, for crossborder identity verification of citizens. The bridge system will act as the connector to the national identification system in the countries involved in the bilateral or multilateral arrangement. Businesses seeking to verify the identity of citizens from partner countries can integrate into their national identification body's verification system. The business entity will send a verification request for non-citizens to the non-citizen's country's identification issuing body through their national connector. The response from the non-citizen's country's identification issuing body will be relayed to the national connector in the business entity's country and then transmitted to the business through their national identification issuing body. The oversight of the bridge system in Ghana could be managed by the National Information Technology Agency, and similar agencies in the counterpart countries. In Ghana, the National Information Technology Agency is responsible for the development of Information Technology architecture and interoperability standards for government applications and networks. This national bridge system could be a proxy model utilised for a continent-wide national identity data exchange by the African Union for all national identification issuing bodies in member states. This initiative would enable businesses to verify the identity of non-citizens in their country by integrating with their national identification issuing bodies.

6.2 Civil Society

1. Implement advocacy and sensitisation programmes

The pivotal role of civil society organisations in shaping public policies and laws, and in holding governments accountable for their implementation, cannot be overstated. They serve as both guardians and partners for the betterment of society. Numerous policies and laws have been shaped by the resolute advocacy of civil society organisations, with the Right To Information law being particularly an example. Civil society organisations have played a crucial role in raising public awareness about this law. Similarly, they can advocate for the government to develop and implement a data governance framework. Given that data protection and privacy are significant concerns for a majority of the surveyed data subjects, civil society organisations can collaborate with partners to educate data subjects about their rights under existing laws and hold data controllers accountable by monitoring and publicising breaches. It's important to ensure that data subjects in rural areas of Ghana are not overlooked, particularly considering the concentration of civil society organisations in urban areas. Civil society organisations focused on digital rights, data governance, and data privacy can offer valuable technical expertise to the Ministry of Communication in developing a data governance framework that aligns with the African Union Data Policy.

2. Create a coalition to advance the cause

One major challenge that civil society in Ghana faces is finance, with most relying on donor funding to implement their initiatives. With this source increasingly becoming limited, it is important to find new ways to develop and fund advocacy and sensitisation programmes on data governance. Consequently, it is recommended that civil society organisations with similar interests in data governance and its related themes join forces to generate and sustain the necessary momentum required to achieve the desired results from their advocacy and sensitisation initiatives for the government to develop and implement a data governance framework that enables cross-border data flow, particularly for identity verification under bilateral arrangements. This can also be used as a tool for coordinated actions across several African countries to accelerate the achievement of the objectives of the African Union Data Policy

6.3 Academia

1. Conduct further research on the topic

It is crucial to further research this topic to determine the most appropriate data governance operating model for Ghana. This model should support the seamless integration and interoperability of identification systems within Ghana's data governance framework. The results will serve as the foundation for civil society organisations to interact with the government, presenting a compelling argument for cross-border data flow, especially for the cross-border verification of Ghanaians to improve access to services and movement.

7.0 Conclusion

Cross-border data flow is critical to the realisation of the objectives of Agenda 2063 with implications for continent-wide economic growth and prosperity, the free movement of people, goods, and services, and the integration of the African economy. As a significant political and socio-economic player, the need to assess the impact of Ghana's data governance and digital systems on cross-border data flow can not be overemphasised.

A review of the Ghana Government Enterprise, e-Government Interoperability, and Data Sharing Policy found a non-alignment with the AU Data Policy. The country's identification systems, both foundation and functional, are not interoperable. The type of integration that exists is data integration, and this does not enable system-level interoperability. Data of the functional identification issuers are simply consolidated into the system of the foundational identification issuer. None of the identification systems is integrated into any external system to enable cross-border data flow. Data privacy and protection is a major concern for data subjects of the Personal Identification issuers in Ghana.

It is recommended that a data governance framework that aligns with the AU Data Policy be developed and adopted. The status of the National Information Technology Agency be elevated to an Authority for the effective implementation of Ghana's data governance framework. To enable crossborder verification of citizens, it is proposed that a bridge system via the National Information Technology Agency to the foundational identification system be created for integration by countries under a bilateral or multilateral arrangement. At the continent-wide level, this can be leveraged to create an organisation similar in nature to the Pan-African Payment and Settlement System, except that, in this case, it would involve connecting the foundation identification systems of member states via a proxy model. Efforts towards achieving the objective of the AU Data Policy must accompanied by, advocacy and sensitisation through coordinated initiatives. To ensure that the government's actions are data-driven, further research is recommended to establish a suitable data governance operating model for the data government framework and to build a compelling business case for cross-border data flow.

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